

Agentic AI System Design Report

AI Career Advisor - ReAct Agent with RAG and Tool Use

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1. Overview

The AI Career Advisor is a single-agent ReAct system that takes a candidate's resume text or skill description as input and autonomously orchestrates a multi-step career guidance workflow.

The system matches the candidate against five AI job market archetypes derived from Projects 3 and 4, retrieves semantically similar job postings via a FAISS vector index, generates a personalised recommendation and assembles a salary-informed career report - all without manual intervention between steps. The project represents the synthesis of the full analytical pipeline built across the course series, demonstrating how individual ML components can be composed into a reasoning agent capable of end-to-end task completion.

Agentic AI systems are distinguished from standard LLM applications by their ability to perceive their environment, reason about it and take actions to achieve goals (Russell & Norvig, 2022). The AI Career Advisor embodies this definition: it perceives the candidate's input, reasons about skill profiles and archetype similarity and acts through a sequence of tool calls to produce a structured career report.

2. System Architecture and Design

2.1 Reasoning loop

The agent implements the ReAct pattern introduced by Yao et al. (2022), which interleaves reasoning traces (Thought) with tool invocations (Action) and their results (Observation). Each cycle produces an explicit log entry, making the agent's decision process fully traceable. The reasoning loop runs sequentially - each step's observation informs the next thought - and terminates when the career report is assembled or a safeguard condition is triggered.

2.2 Memory and state

The agent maintains a single state dictionary that persists across all steps within a session, accumulating extracted skills, archetype match, RAG context, generated recommendation, live job postings and salary data. There is no cross-session memory - each run starts fresh. This design prioritises simplicity and privacy: candidate data is never persisted between sessions, which is appropriate for a career guidance tool where session isolation reduces data exposure risk.

2.3 Tool registry

Six tools are registered and callable by the agent: `extract_skills()` applies the 50-skill taxonomy from Project 3; `match_archetype()` computes cosine similarity against five archetype centroids from Project 4; `rag_retrieve()` queries a FAISS index of 1,088 European job postings from Project 4 using DistilBERT embeddings; `generate_recommendation()` produces structured template-based career guidance conditioned on fit level and RAG context; `fetch_live_jobs()` calls the Adzuna Jobs API for current postings matching the archetype; and `get_salary_range()` returns DACH salary statistics from Project 2. Each tool has a defined input/output contract and operates independently of the others.

2.4 RAG implementation

Retrieval-Augmented Generation (RAG) grounds the agent's recommendations in real job market evidence rather than relying solely on template text. Job descriptions from Project 4 are encoded as 768-dimensional DistilBERT embeddings and indexed in FAISS (Johnson et al., 2021). At query time, the candidate's skill profile is embedded and the top-k most semantically similar postings are retrieved, filtered by role category and passed as context to the recommendation step. This follows the RAG architecture described by Lewis et al. (2020), combining retrieval of relevant documents with generation conditioned on that context.

3. Decision Logic and Behavior

The agent's decision logic operates at two levels: routing and tool selection. At the routing level, the fit score determines which recommendation template is used - strong (≥ 0.4), partial (≥ 0.2) or none (< 0.2) and whether live job postings are fetched (skipped for no-fit profiles to avoid misleading results). At the tool selection level, the agent follows a fixed sequential pipeline: skills must be extracted before archetypes can be matched, and RAG context must be retrieved before recommendations can be generated. This

deterministic ordering reflects the data dependencies between steps and ensures consistent, reproducible behaviour.

The agent produces meaningfully distinct outputs across all input types. Strong fit outputs confirm the archetype match, reference actual detected skills and provide specific development focus areas. Partial fit outputs identify the closest archetype and name the specific skills required to strengthen the match. No-fit outputs distinguish between candidates with no technical skills (who may benefit from AI applications in their existing field) and candidates with minimal skills (who are directed toward foundational AI learning pathways). This three-tier routing ensures the agent's guidance is contextually appropriate rather than generic.

4. Safety, Reliability and Transparency

4.1 Safeguards

Two explicit safeguard conditions protect against degenerate inputs. The first triggers on inputs shorter than 10 characters, returning an error message and terminating the pipeline immediately. The second triggers when no skills are detected from the taxonomy, returning a guidance message without proceeding to archetype matching or job fetching. Both safeguards are logged in the reasoning trace, maintaining full transparency about why the pipeline was short-circuited.

4.2 Transparency

The reasoning trace - a full log of every Thought-Action-Observation cycle - is included in every career report. This gives candidates visibility into how the recommendation was produced, which tools were called and what each step returned. Transparency in agentic systems is increasingly recognised as a prerequisite for responsible deployment: Wachter et al. (2017) argue that automated decision systems must provide meaningful explanations to affected individuals and the reasoning trace is the AI Career Advisor's mechanism for meeting this standard.

4.3 Reliability

All core tools operate on local models and data, eliminating dependency on external API availability for the primary recommendation output. The Adzuna API call is the only external dependency at runtime - if it fails, the agent continues and omits the live jobs section rather than crashing. The GPT-2 generation step was replaced with a deterministic template approach after hallucinations were observed in testing, prioritising recommendation reliability over generative flexibility.

5. Observed Behavior and Limitations

The agent behaved as designed across all five test scenarios. The most notable finding was Scenario 4 - an automotive engineer with Python and computer vision skills correctly matched to the Deep Learning & AI Research archetype (score=0.5, strong) despite having no formal AI job title. This validates the core value proposition of skill-based matching over label-based categorisation, consistent with the findings of Projects 3 and 4.

Three limitations were observed in execution. First, the live job fetch returns repeated results across scenarios because the Adzuna query is built from a fixed archetype keyword rather than the candidate's specific skill profile - the same NVIDIA Computer Vision postings appeared for all Deep Learning archetype matches. Second, RAG retrieval returns only 1–2 results for some archetypes due to category imbalance in the FAISS index, where AI_Engineering postings dominate over Data_Analytics and AI_Integration. Third, the 50-skill taxonomy produces sparse extraction on non-technical resumes - a finance analyst with strong Excel skills receives no skill match for Excel since it is absent from the taxonomy, potentially underestimating their analytical capability.

6. Ethical and Responsible Use Considerations

The most significant ethical concern is the risk of the system being repurposed as a hiring screening tool rather than a candidate-facing career guidance aid. If recruiters used archetype similarity scores to filter or rank applicants, the system would encode and amplify the biases present in the underlying job market data - which was collected primarily from UK and German Adzuna postings and therefore reflects those markets' specific hiring norms, language and role structures. Candidates from different educational backgrounds, non-English-speaking countries or career paths not well-represented in the Adzuna index may be systematically disadvantaged even when they possess equivalent competencies described in different terminology.

A second concern relates to the autonomy of the recommendation. The agent produces confident-sounding career guidance - 'Your profile aligns strongly with...' - without any uncertainty quantification or acknowledgement that the assessment is based on a narrow keyword taxonomy. A candidate receiving a no-fit response may be discouraged from pursuing AI roles despite having relevant experience that falls outside the taxonomy. Jobin et al. (2019), in their review of global AI ethics guidelines, identify transparency, non-maleficence and human oversight as the most consistently cited principles - all three are relevant here. The system addresses transparency through the reasoning trace but does not currently provide uncertainty estimates or encourage human review of its outputs. Under the EU AI Act, any deployment of this system in a recruitment context would require classification as a high-risk AI system under Annex III,

triggering requirements for conformity assessment, human oversight and bias monitoring. The current implementation is designed exclusively for candidate-facing career orientation and explicitly excludes any recruiter-facing functionality.

7. Future Improvements

The most impactful near-term improvement would be replacing the fixed archetype keyword query in `fetch_live_jobs()` with a dynamic query built from the candidate's actual detected skills.

This would produce personalised live job results rather than the same archetype-level results for every candidate matching a given persona.

Replacing the 50-skill keyword taxonomy with a named entity recognition model fine-tuned on resume text would significantly improve skill extraction coverage, particularly for non-technical profiles and for skills expressed in non-standard or multilingual terminology. This would directly reduce the false negative rate that currently causes some analytically capable candidates to receive no-fit responses.

Adding uncertainty quantification to the archetype match - for example, reporting the similarity scores for all five archetypes rather than only the best match - would allow candidates to understand cases where their profile sits between archetypes. This would improve the interpretability of partial-fit responses and reduce the risk of candidates acting on an artificially confident recommendation. Finally, extending the system to a multi-session memory architecture using a lightweight vector store would allow the agent to track skill development over time and provide longitudinal career guidance rather than one-off snapshots.

8. References

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